

Project idea: Two-Player Reaction Time Matching Game using PICMicro 16F877A

We would like to build a PIC Assembly code on a 16F877A microcontroller to create an interactive microcontroller-based reaction game for two players that employs push buttons, visual displays, a 16X2 LCD and LEDs. A central judge or referee on the 16F877A microcontroller initiates a round through a push button, which generates a random time value in the range [0 ... 60] seconds. The value should be displayed on the 16X2 LCD.

Each player is equipped with a distinct physical hardware station featuring a push button and a 4-digit 7-segment display showing independent timers.

The core game logic involves precise time measurement. Once the delay value is displayed on the 16X2 LCD and a master push button is clicked, players individual timers start and players must stop them as close to the target duration selected by the judge as possible. Upon determining the winner, the system checks whose final time is closest to the fixed target timer and blinks the corresponding physical indicator LED on that player's side. A particular player timer can be slightly more or less than the judge's selected time. However, in absolute values, the winner has a closer selected timer value to the judge's selected value.

Description

Hardware components:

- 16F877A microcontroller, breadboard, 2 different-colored LED indicators, resistors (220, 4.7K, 10K), 3 push buttons, 2 visual displays (4-digit 7-segments), 16X2 LCD.

Player Hardware Stations:

- **Input:** Two high-durability push buttons tied to hardware interrupt pins (GPIO)..
- **Visual Displays:** Two discrete 4-digit 7-segment displays mapped via synchronous protocols to display running timers down to millisecond precision.
- **Indicator LEDs:** Two dedicated physical status LEDs mapped to designated GPIO lines to dynamically indicate the winning player.

Simulation Flow:

1. Initially, the 16F877A microcontroller acts as the judge and selects a random number in the range [0 ... 60] upon clicking a master push button. The selected time value is displayed on the 16X2 LCD so the 2 players can see the value.

2. The 16F877A microcontroller turns OFF the 2 LED indicators on the breadboard. During that time the players get ready by each placing his/her thumb on one of the 2 push buttons. If push button clicks occur during that time, the clicks are ignored (push buttons are disabled).
3. When the master push button is clicked, the 2 indicator LEDs on the breadboard go ON for 1 second and then go OFF.
4. When the 2 indicator LEDs go OFF, the 2 visual displays (4-digit 7-segment) start incrementing time from 00.00 up to the value that has been selected by the judge (16F877A microcontroller). The players' push buttons are now enabled.
5. When the time value on both 4-digit 7-segment get near to the value selected by the judge, each player will try to stop his/her timer when the value gets the closest to the judge's time value.
6. When a player clicks on his/her push button, the respective time should stop incrementing on the player's 4-digit 7-segment display.
7. When the 2 players click on their respective push buttons, the 16F877A microcontroller must decide who among the 2 players is the winner. The winner must have the smallest delta (absolute value of the judge's time minus the player's timer value).
8. The 16F877A microcontroller blinks the LED of the winner 3 times (0.25 sec ON, 0.25 sec OFF).
9. The 16F877A microcontroller must keep the score of the 2 players and display it on the 16X2 LCD.
10. A new round is initiated by going to point 1 above.
11. The game ends when any of the 2 players score gets to 5 first.

Hint: You should handle edge cases, like what if one of the players does not click on his/her push button during a round. Consider other edge cases as well.

What you should do:

- Use proteus application to build the schematic for the whole system, including the 16F877A microcontroller, the 3 push buttons (1 master, 2 for the players), 16X2 LCDs, the LEDs, etc.
- Use external components such transistors and multiplexers if needed.
- Remember to add a 10K Ω pull-up resistors to the push buttons (or use the internal weak pull-up resistors of PortB), add a 4MHZ oscillator (with $2 \times 15\text{pF}$ capacitors) and a 10K Ω pull-up resistor to the MCLR pin.
- Enable interrupts on the push buttons (if you wish to use interrupts).
- Build the PIC assembly code for the 16F877A microcontroller that implements the behavior described above under MPLAB IDE.
- Assemble the project and make sure you get a successful build. Use the simulator if you wish to make sure the behavior is correct.
- Send the zipped folder that contains the MPLAB code for the project before the deadline. If the deadline is reached and you are still having problems with your code, just send it as is!